



KTU NOTES

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**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

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magnitude of output signal to the input signal and phase between the signals disappeared. Hence frequency response tests are not generally carried out for systems with constant inputs.

The transfer function of a system can be described in frequency domain by the following equation:

$$G(j\omega) = G(s) \Big|_{s=j\omega}$$

The closed loop transfer function in frequency domain is

$$\frac{C(j\omega)}{R(j\omega)} = M(j\omega) = |M(j\omega)| \angle M(j\omega)$$

$G(j\omega)H(j\omega)$ = open loop transfer function in frequency domain

FREQUENCY DOMAIN SPECIFICATIONS

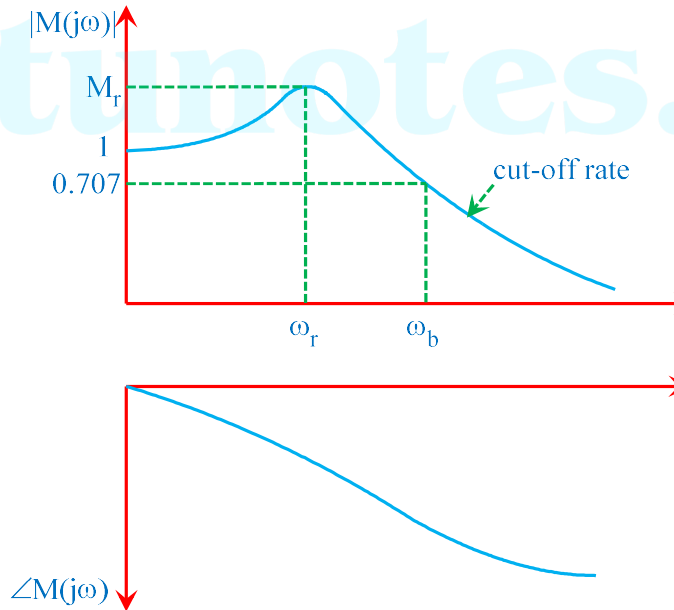
RESONANT PEAK (M_r)	It is the maximum value of magnitude of closed-loop transfer function $ M(j\omega) $. M_r gives indication on the relative stability of a stable system. Large M_r corresponds to a large maximum overshoot.
RESONANT FREQUENCY (ω_r)	It is the frequency at which resonant peak occurs. ω_r gives an indication of the speed of transient response.
BANDWIDTH (ω_b)	For feedback control systems, the range of frequencies over which the magnitude of the closed-loop transfer function is equal to or greater than -3dB. Large BW corresponds to small rise time. BW also gives an indication of the noise-filtering capability and robustness of the system. (A robust system is one that is insensitive to parameter variations)
CUT-OFF RATE	The slope of $ M(j\omega) $ near the -3dB cut-off frequency is called the cut-off rate. Bandwidth alone is inadequate to indicate the ability of a system to distinguish signals from noise. Two systems may have the same bandwidth, but the cut-off rates may be different.
GAIN MARGIN (GM)	It is the amount of gain in dB that can be added to the open-loop system before it becomes unstable.

Gain cross-over frequency is the frequency at which the function is unity (or the frequency at which the gain

$$\text{Phase margin} = 180^\circ + \phi_{gc}$$

$$\phi_{gc} = \angle G(j\omega_{gc})$$

Typical gain – phase characteristics of a feedback control system is shown below



EXPRESSION FOR RESONANT FREQUENCY

The closed loop transfer function of a standard second-order system is

$$M(s) = \frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The sinusoidal transfer function is given by

$$4u(1-u^2) = (4\xi u) \times 2\xi$$

$$1-u^2 = 2\xi^2$$

$$u^2 = 1 - 2\xi^2$$

$$\left(\frac{\omega_r}{\omega_n}\right)^2 = 1 - 2\xi^2$$

$$\omega_r = \omega_n \sqrt{1 - 2\xi^2}$$

EXPRESSION FOR RESONANT PEAK

At resonant frequency, $|M(j\omega)| = M_r$

$$|M(j\omega)| = \frac{1}{\sqrt{(1-u^2)^2 + (2\xi u)^2}}$$

$$\text{Resonant peak, } M_r = \frac{1}{\sqrt{(1-(1-2\xi^2))^2 + (2\xi(\sqrt{1-2\xi^2}))^2}} = \frac{1}{\sqrt{4\xi^4 + 4\xi^2}}$$

$$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

EXPRESSION FOR BAND WIDTH

$$\text{At cut-off frequency, } |M(j\omega)| = \frac{1}{\sqrt{(1-u^2)^2 + (2\xi u)^2}} = \frac{1}{\sqrt{2}}$$

$$(1-u^2)^2 + (2\xi u)^2 = 2$$

$$1 - 2u^2 + u^4 + 4\xi^2 u^2 = 2$$

$$u^4 - 2(1-2\xi^2)u^2 - 1 = 0$$

$$u^2 = (1-2\xi^2) \pm \sqrt{(1-2\xi^2)^2 + 1} = (1-2\xi^2) \pm \sqrt{4\xi^4 - 4\xi^2 + 2}$$

$$\text{When } |M(j\omega)| = \frac{1}{\sqrt{2}}, \omega = \omega_b$$

Note 2	When $\xi > 0.707$, there is no resonant peak and $\omega_r = 0$.
Note 3	When $0 < \xi < 0.707$, $M_r > 1$ and $\omega_r < \omega_n$.
Note 4	When $\xi = 0$, $M_r = \infty$ and $\omega_r = \omega_n$.
Note 5	BW decreases as ξ increases.

Find the resonant peak and resonant frequency of a unity feedback system with

$$M(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{100}{s^2 + 10s + 100}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\omega_n = 10 \text{ rad/sec}$$

$$\xi = 0.5$$

$$\text{Resonant peak, } M_r = \frac{1}{2\xi\sqrt{1-\xi^2}} = \frac{1}{2 \times 0.5\sqrt{1-0.5^2}} = 1.154$$

$$\text{Resonant frequency, } \omega_r = \omega_n\sqrt{1-2\xi^2} = 10 \times \sqrt{1-2 \times 0.5^2} = 7.071 \text{ rad/sec}$$

Find the resonant peak, resonant frequency and bandwidth of a unity feedback system with

$$G(s) = \frac{36}{s(s+8)}.$$

$$M(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{36}{s^2 + 8s + 36}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\omega_n = 6 \text{ rad/sec}$$

$$e^{-\sqrt{1-\xi_1^2}} = 0.5$$

$$-\frac{\xi_1 \pi}{\sqrt{1-\xi_1^2}} = -0.6931$$

$$\xi = 0.2154$$

$$f_d = \frac{1}{0.2} = 5 \text{ Hz}$$

$$\omega_d = 2 \times \pi \times 5 = 31.41 \text{ rad/sec}$$

$$\omega_d = \omega_n \sqrt{1-\xi^2} = 31.41$$

$$\omega_n = 32.16 \text{ rad/sec}$$

$$\text{Resonant peak, } M_r = \frac{1}{2\xi\sqrt{1-\xi^2}} = 2.377$$

$$\text{Resonant frequency, } \omega_r = \omega_n \sqrt{1-2\xi^2} = 30.64 \text{ rad/sec}$$

$$\text{Bandwidth, } \omega_b = \omega_n \sqrt{(1-2\xi^2) \pm \sqrt{4\xi^4 - 4\xi^2 + 2}} = 48.3 \text{ rad/sec}$$

3. Poles or zeros on the real axis
4. Complex conjugate poles

BASIC FACTORS of $G(j\omega)$

1	Constant Gain K $G(s) = K$
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$$G(j\omega) = K$$

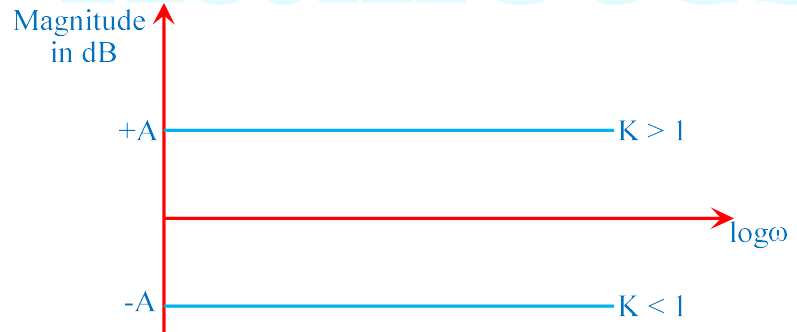
$$20 \log |G(j\omega)| = 20 \log K = A ; \text{ has constant magnitude}$$

$$\text{Phase angle, } \phi = \angle G(j\omega) = 0$$

When $K > 1$, A is positive

When $K = 1$, $A = 0$

When $K < 1$, A is negative



2	$G(s) = \frac{1}{s}$
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$$G(j\omega) = \frac{1}{j\omega}$$

$$20 \log |G(j\omega)| = A = 20 \log \frac{1}{\omega} = -20 \log \omega$$

$$\text{Phase angle, } \phi = \angle G(j\omega) = -90^\circ$$

When $\omega = 0.1$, $A = 20$ dB

When $\omega = 1$, $A = 0$ dB

When $\omega = 10$, $A = -20$ dB

When $\omega = 100$, $A = -40$ dB

3

$$G(s) = \frac{1}{s^n}$$

$$G(j\omega) = \frac{1}{(j\omega)^n}$$

$$20 \log |G(j\omega)| = A = 20 \log \frac{1}{\omega^n} = -20n \log \omega$$

$$\text{Phase angle, } \phi = \angle G(j\omega) = -n90^\circ$$

A Vs log ω is a straight line with slope equal to -20ndB/decade.

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$$G(s) = s$$

$$G(j\omega) = j\omega$$

$$20 \log |G(j\omega)| = A = 20 \log \omega$$

$$\text{Phase angle, } \phi = \angle G(j\omega) = +90^\circ$$

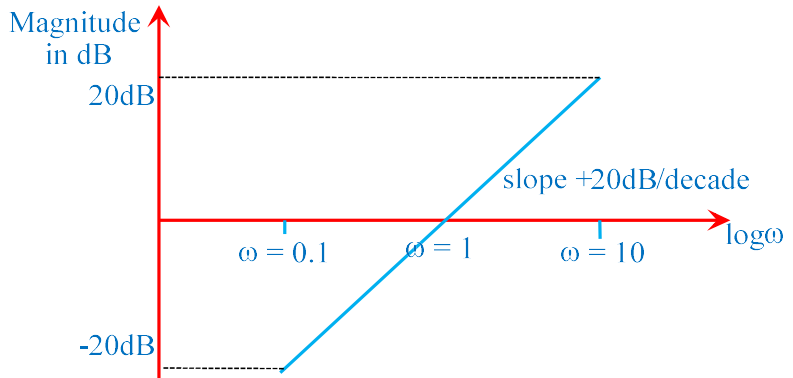
When $\omega = 0.1$, $A = -20$ dB

When $\omega = 1$, $A = 0$ dB

When $\omega = 10$, $A = +20$ dB

When $\omega = 100$, $A = +40$ dB

A Vs log ω is a straight line with slope equal to +20dB/decade.



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$$G(s) = s^n$$

$$\sqrt{1 + \omega^2 T^2}$$

Case 1: $\omega T \ll 1$ (for low frequency)

$\omega^2 T^2$ is neglected

$$A = -20 \log 1 = 0$$

Case 2: $\omega T \gg 1$ (for high frequency)

$$A = -20 \log \omega T$$

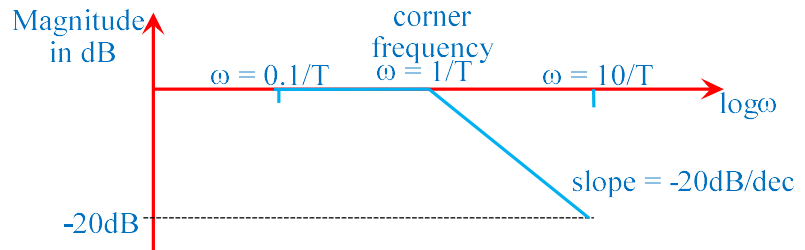
(When $\omega T = 1$, $A = 0$ dB and when $\omega T = 10$, $A = -20$ dB. Hence, A Vs $\log \omega$ with slope -20 dB/decade)

Magnitude plot is a horizontal line of 0 dB when $\omega T \ll 1$.

Magnitude plot is a straight line with slope -20 dB/decade when $\omega T \gg 1$.

These two straight lines meet at a point $\omega = \frac{1}{T}$ which is called **corner frequency**

Note: Bode plots constructed using straight lines are called asymptotic (approximate) plots are not correct at or near corner frequencies.



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$$G(s) = \frac{1}{(1 + sT)^m}$$

$$G(j\omega) = \frac{1}{(1 + j\omega T)^m}$$

$$\angle G(j\omega) = -m \tan^{-1} \omega T$$

$$A = 20 \log |G(j\omega)| = -20m \log \sqrt{1 + \omega^2 T^2}$$

Case 1: $\omega T \ll 1$ (for low frequency)

-40dB

slope = -40dB/dec

8

$$G(s) = 1 + sT$$

$$G(j\omega) = 1 + j\omega T$$

$$|G(j\omega)| = \sqrt{1 + \omega^2 T^2}$$

$$\angle G(j\omega) = \tan^{-1} \omega T$$

$$A = 20 \log |G(j\omega)| = 20 \log \sqrt{1 + \omega^2 T^2}$$

Case 1: $\omega T \ll 1$ (for low frequency)

$\omega^2 T^2$ is neglected

$$A = 20 \log 1 = 0$$

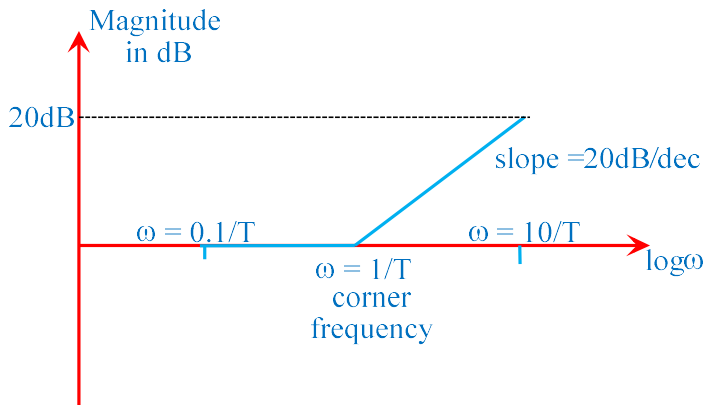
Case 2: $\omega T \gg 1$ (for high frequency)

$$A = 20 \log \omega T$$

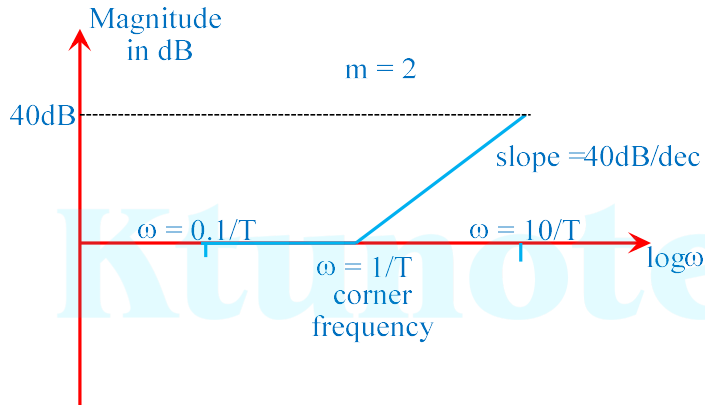
(When $\omega T = 1$, $A = 0$ dB and when $\omega T = 10$, $A = 20$ dB. Hence, A Vs $\log \omega$ slope 20dB/decade)

Magnitude plot is a horizontal line of 0dB when $\omega T \ll 1$.

Magnitude plot is a straight line with slope 20dB/decade when $\omega T \gg 1$.



Magnitude plot is a straight line with slope 20dB/decade when $\omega T \gg 1$.



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$$G(s) = \frac{1}{s^2 T^2 + 2\xi T s + 1}$$

$$G(j\omega) = \frac{1}{-\omega^2 T^2 + j2\xi\omega T + 1}$$

$$|G(j\omega)| = \frac{1}{\sqrt{(1 - \omega^2 T^2)^2 + 4\xi^2 \omega^2 T^2}}$$

$$\angle G(j\omega) = -\tan^{-1} \frac{2\xi\omega T}{1 - \omega^2 T^2}$$

$$A = 20 \log |G(j\omega)| = 20 \log \frac{1}{\sqrt{(1 - \omega^2 T^2)^2 + 4\xi^2 \omega^2 T^2}}$$

Case 1: $\omega T \ll 1$ (for low frequency)

$\omega^2 T^2$ is neglected

$$A = -20 \log 1 = 0$$

Case 2: $\omega T \gg 1$ (for high frequency)

$$(1 - \omega^2 T^2)^2 \approx \omega^4 T^4$$

Construct the Bode plot for a unity feedback control system having, $G(s) = \frac{2000}{s(s+1)(s+100)}$

$$G(s) = \frac{2000}{s(s+1)(s+100)}$$

Converting to time-constant form,

$$G(s) = \frac{20}{s(1+s)(1+0.01s)}$$

$$\text{Sinusoidal transfer function, } G(j\omega) = \frac{20}{j\omega(1+j\omega)(1+j0.01\omega)}$$

Factor	Corner frequency rad/s	slope	Net slope	Frequency
K = 20	-	0	0	
$\frac{20}{j\omega}$	-	-20	-20	$\omega = 0.5 \text{ rad/s}$ $A = 20\log 20$ $A = 32 \text{ dB}$
$\frac{1}{1+j\omega}$	1	-20	-40	$\omega = 1 \text{ rad/s}$
$\frac{1}{1+j0.01\omega}$	100	-20	-60	$\omega = 100 \text{ rad/s}$

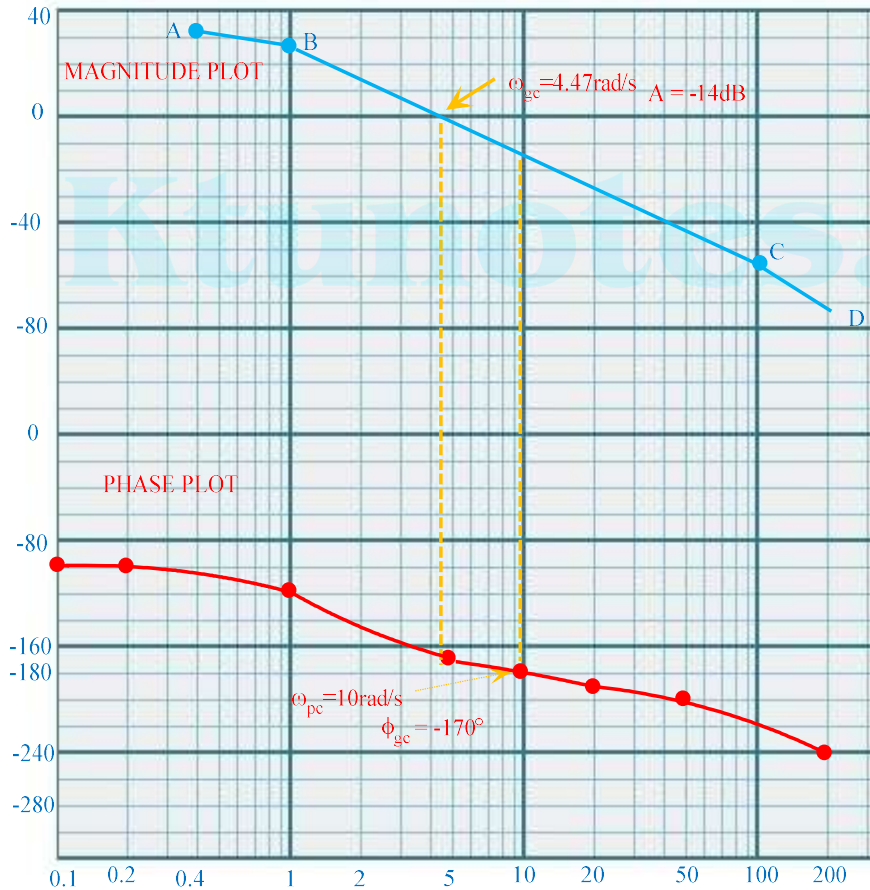
Note: In second column, the corner frequency should be arranged in increasing order.

Recheck:-

$\omega \text{ rad/s}$	A in dB
$\omega = 0.5$	$A = 20\log 20 - 20\log \omega = 32 \text{ dB}$
$\omega = \omega_{c1} = 1$	$A = 20\log 20 - 20\log \omega = 26 \text{ dB}$
$\omega = \omega_{c1} = 100$	$A = 20\log 20 - 20\log \omega - 20\log \omega = -54 \text{ dB}$

Phase Margin $PM = 180 + \phi_{gc} = 180 - 170 = 10^\circ$

Both GM and PM are positive (or $\omega_{gc} < \omega_{pc}$). The system is absolutely stable.



Sketch the Bode plot of the open loop transfer function, $G(s) = \frac{10}{s(1+0.5s)(1+...)}$

$\frac{1}{1 + j0.1\omega}$	10	-20	-60	$\omega = 10$ rad/s
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Note: In second column, the corner frequency should be arranged in increasing order.

Recheck:-

ω rad/s	A in dB
$\omega = 1$	$A = 20\log 10 - 20\log \omega = 20\text{dB}$
$\omega = \omega_{c1} = 2$	$A = 20\log 10 - 20\log \omega = 14\text{dB}$
$\omega = \omega_{c1} = 10$	$A = 20\log 10 - 20\log \omega - 20\log 0.5\omega = -14\text{dB}$
$\omega = \omega_{c1} = 50$	$A = 20\log 10 - 20\log \omega - 20\log 0.5\omega - 20\log 0.1\omega = -24\text{dB}$
$\omega = \omega_{gc}, A = 0$	$20\log 10 - 20\log \omega - 20\log 0.5\omega = 0; \omega_{gc} = 4.47$ rad/s

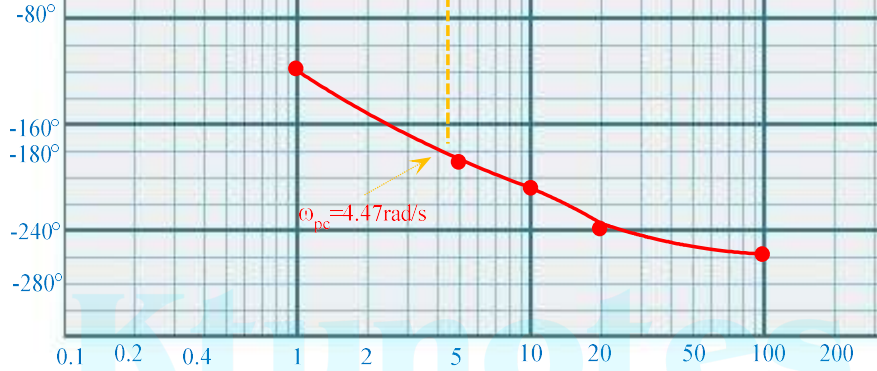
$$\phi = \angle G(j\omega) = -90 - \tan^{-1} 0.5\omega - \tan^{-1} 0.1\omega$$

ω	1	5	10	20	50	100	200	∞
ϕ	-122	-185	-214	-238	-257	-264	-267	-270

At $\omega = \omega_{pc}$, $\phi = -180$; $\phi = \angle G(j\omega) = -90 - \tan^{-1} 0.5\omega - \tan^{-1} 0.1\omega = -180$

Note: Use CALC button of fx-991ES or EX Casio calculator to find ϕ in one step.

Here, $\omega_{gc} = \omega_{pc} = 4.47\text{rad/s}$, hence the system is marginally stable.



Draw the Bode pot of the open loop transfer function, $G(s) = \frac{36(1+0.2s)}{s^2(1+0.05s)(1+0.01s)}$

$$G(s) = \frac{36(1+0.2s)}{s^2(1+0.05s)(1+0.01s)}$$

$$\text{Sinusoidal transfer function, } G(j\omega) = \frac{36(1+j0.2\omega)}{(j\omega)^2(1+j0.05\omega)(1+j0.01\omega)}$$

Factor	Corner frequency rad/s	slope	Net slope	Frequency
$K = 36$	-	0	0	
$\frac{36}{(j\omega)^2}$	-	-40	-40	$\omega = 2 \text{ rad/s}$ $A = 20 \log 36$ $A = 19$
$1 + j0.2\omega$	5	20	-20	$\omega = 5 \text{ rad/s}$
$\frac{1}{1 + j0.05\omega}$	20	-20	-40	$\omega = 20 \text{ rad/s}$
$\frac{1}{1 + j0.01\omega}$	100	-20	-60	$\omega = 100 \text{ rad/s}$



At $\omega = \omega_{pc} = 37.4$ rad/s, $A = 20\log 36 - 40\log \omega + 20\log 0.2\omega - 20\log 0.05\omega = -19.8$ dB

At $\omega = \omega_{gc} = 7.2$ rad/s, $\phi = -180 + \tan^{-1} 0.2\omega - \tan^{-1} 0.05\omega - \tan^{-1} 0.01\omega = -149^\circ$

Gain Margin $GM = -A$ at $\omega_{pc} = +19.8$ dB

Phase Margin $PM = 180 + \phi_{gc} = 180 - 149 = 31^\circ$

Both GM and PM are positive (or $\omega_{gc} < \omega_{pc}$). The system is absolutely stable.

$\frac{0.15}{j\omega}$	-	-20	-20	$\omega = 0.1 \text{ rad/s}$ $A = 20\log 0.15$ $A = 3.5 \text{ dB}$
$\frac{1}{1 + j0.5\omega}$	2	-20	-40	$\omega = 2$
$1 + j0.33\omega$	3	20	-20	$\omega = 3 \text{ rad/s}$
$\frac{1}{1 + j0.04\omega - 0.01\omega^2}$ $T^2 = 0.01; T = 0.1$	10	-40	-60	$\omega = 10$

Recheck:-

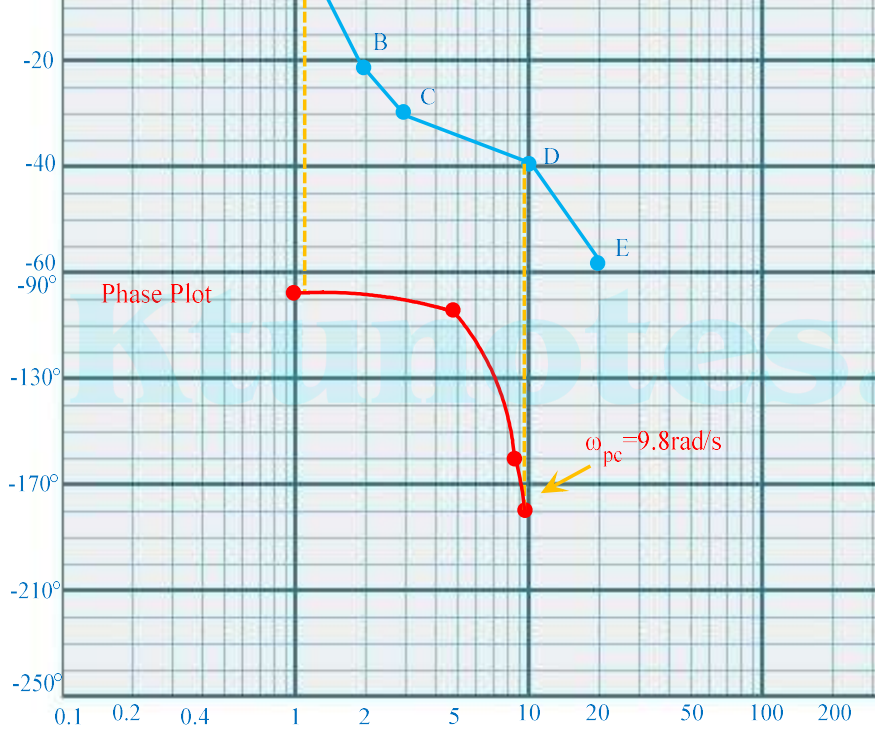
$\omega \text{ rad/s}$	A in dB
$\omega = 1$	$A = 20\log 0.15 - 20\log \omega = 3.5 \text{ dB}$
$\omega = \omega_{c1} = 2$	$A = 20\log 0.15 - 20\log \omega = -22 \text{ dB}$
$\omega = \omega_{c1} = 3$	$A = 20\log 0.15 - 20\log \omega - 20\log 0.5\omega =$
$\omega = \omega_{c1} = 10$	$A = 20\log 0.15 - 20\log \omega - 20\log 0.5\omega + 20\log 0 =$
$\omega = \omega_{c1} = 20$	$A = 20\log 0.15 - 20\log \omega - 20\log 0.5\omega + 20\log 0.33\omega -$
$\omega = \omega_{gc}, A = 0$	$20\log 0.15 - 20\log \omega = 0; \omega_{gc} = 0.15 \text{ rad/s}$

$$\phi = \angle G(j\omega) = -90 - \tan^{-1} 0.5\omega + \tan^{-1} 0.33\omega - \tan^{-1} \frac{0.04\omega}{1 - 0.01\omega^2}$$

Note: For $\omega < 10$, use equation $\phi = -90 - \tan^{-1} 0.5\omega + \tan^{-1} 0.33\omega - \tan^{-1} \frac{0.04\omega}{1 - 0.01\omega^2}$

For $\omega > 10$, use equation $\phi = -90 - \tan^{-1} 0.5\omega + \tan^{-1} 0.33\omega - 180 - \tan^{-1} \frac{0.04\omega}{1 - 0.01\omega^2}$

Remember:		
Quadrant 1	$1 + j\omega$	$\phi = \tan^{-1} \omega$
Quadrant 2	$-1 + j\omega$	$\phi = 180 - \tan^{-1}(-\omega)$
Quadrant 3	$-1 - j\omega$	$\phi = 180 - \tan^{-1}(\omega)$



At $\omega = \omega_{pc} = 9.8 \text{ rad/s}$, $A = 20\log 0.15 - 20\log \omega - 20\log 0.5\omega + 20\log 0.33\omega = -40 \text{ dB}$

At $\omega = \omega_{gc} = 0.15 \text{ rad/s}$, $\phi = -90 - \tan^{-1} 0.5\omega + \tan^{-1} 0.33\omega - \tan^{-1} \frac{0.04}{1 - 0.04\omega} = -92^\circ$

Gain Margin $GM = -A$ at $\omega_{pc} = +40 \text{ dB}$

Phase Margin $PM = 180 + \phi_{gc} = 180 - 92 = 88^\circ$

Both GM and PM are positive (or $\omega_{gc} < \omega_{pc}$). The system is absolutely stable.

Draw the Bode plot of the open loop transfer function, $G(s) = \frac{25(s+1)}{s(s+2)(s^2+2s+2)}$

POLAR PLOT

A sinusoidal transfer function $G(j\omega)$ having magnitude $|G(j\omega)|$ and phase angle on polar co-ordinates varying ω from 0 to ∞ .

Polar plot is the locus of phasor $|G(j\omega)|\angle G(j\omega)$ as ω is varied from zero to ∞ .

The polar plot is very useful in determining the stability of a closed loop system frequency response.

Note: For closed-loop system stability, the open-loop transfer function is $G(s)H(s)$.

PROCEDURE TO SKETCH THE POLAR PLOT

1	Determine the transfer function $G(s)$ of the system
2	Put $s = j\omega$ in the transfer function to obtain $G(j\omega)$
3	Calculate the magnitudes of $G(j\omega)$ at limiting frequencies of $\omega \rightarrow 0$ and $\omega \rightarrow \infty$
4	Calculate the phase angles of $G(j\omega)$ at limiting frequencies of $\omega \rightarrow 0$ and $\omega \rightarrow \infty$
5	Calculate the value of ω at which the polar plot crosses the real axis and the magnitude of $G(j\omega)$ at the point.
6	Calculate the value of ω at which the polar plot crosses the imaginary axis and the magnitude of $G(j\omega)$ at the point.
7	Sketch the polar plot

1	Polar plot of $G(s) = \frac{1}{s}$			
	$G(j\omega) = \frac{1}{j\omega}$			
	$ G(j\omega) = \frac{1}{\omega}$			
	$\angle G(j\omega) = -90^\circ$			
	ω	$ G(j\omega) $	$\angle G(j\omega)$	
	0	∞	-90°	
	∞	0	-90°	
2	Polar plot of $G(s) = s$			

$$|G(j\omega)| = \frac{1}{\omega^2}$$

$$\angle G(j\omega) = -180^\circ$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	∞	-180°
∞	0	-180°



4

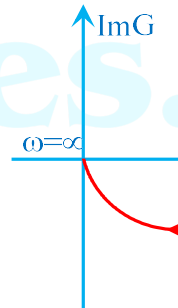
$$\text{Polar plot of } G(s) = \frac{1}{1+sT}$$

$$G(j\omega) = \frac{1}{1+j\omega T}$$

$$|G(j\omega)| = \frac{1}{\sqrt{1+\omega^2 T^2}}$$

$$\angle G(j\omega) = -\tan^{-1} \omega T$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	1	0°
$1/T$	0.707	-45°
∞	0	-90°



5

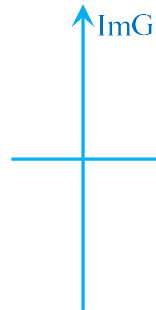
$$\text{Polar plot of } G(s) = 1+sT$$

$$G(j\omega) = 1+j\omega T \text{ (Real part is always 1)}$$

$$|G(j\omega)| = \sqrt{1+\omega^2 T^2}$$

$$\angle G(j\omega) = \tan^{-1} \omega T$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	1	0°
∞	∞	90°



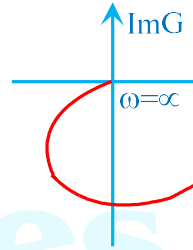
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$$\text{Polar plot of } G(s) = \frac{1}{s(1+sT)}$$

$$G(j\omega) = \frac{1}{j\omega(1+j\omega T)}$$

When $\omega = 0$, real part is -1 and imaginary part is ∞ .

7	Polar plot of $G(s) = \frac{1}{(1+sT_1)(1+sT_2)}$	
	$G(j\omega) = \frac{1}{(1+j\omega T_1)(1+j\omega T_2)}$	
	$ G(j\omega) = \frac{1}{\sqrt{1+\omega^2 T_1^2} \times \sqrt{1+\omega^2 T_2^2}}$	
	$\angle G(j\omega) = -\tan^{-1} \omega T_1 - \tan^{-1} \omega T_2$	
	ω	$\angle G(j\omega)$
	0	0°
	∞	-180°



Sketch the polar plot for the open loop transfer function $G(s) = \frac{50}{(s+1)(s+2)}$

$$G(s) = \frac{25}{(1+s)(1+0.5s)}$$

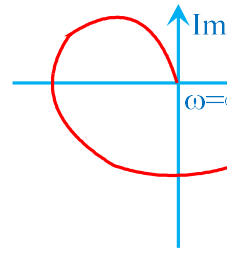
$$G(j\omega) = \frac{25}{(1+j\omega)(1+j0.5\omega)}$$

$$|G(j\omega)| = \frac{25}{\sqrt{1+\omega^2} \times \sqrt{1+(0.5\omega)^2}}$$

$$\angle G(j\omega) = -\tan^{-1} \omega - \tan^{-1} 0.5\omega$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	25	0
∞	0	-180

$(1+sT_1)(1+sT_2)(1+sT_3)$		
$G(j\omega) = \frac{1}{(1+j\omega T_1)(1+j\omega T_2)(1+j\omega T_3)}$		
$ G(j\omega) = \frac{1}{\sqrt{1+\omega^2 T_1^2} \times \sqrt{1+\omega^2 T_2^2} \times \sqrt{1+\omega^2 T_3^2}}$		
$\angle G(j\omega) = -\tan^{-1} \omega T_1 - \tan^{-1} \omega T_2 - \tan^{-1} \omega T_3$		
ω	$ G(j\omega) $	$\angle G(j\omega)$
0	1	0°
∞	0	-270°



Sketch the polar plot for the open loop transfer function $G(s) = \frac{40}{(1+2s)(1+3s)(1+4s)}$

$$G(s) = \frac{40}{(1+2s)(1+3s)(1+4s)}$$

$$G(j\omega) = \frac{40}{(1+j2\omega)(1+j3\omega)(1+j4\omega)}$$

$$|G(j\omega)| = \frac{40}{\sqrt{1+4\omega^2} \times \sqrt{1+9\omega^2} \times \sqrt{1+16\omega^2}}$$

$$\angle G(j\omega) = -\tan^{-1} 2\omega - \tan^{-1} 3\omega - \tan^{-1} 4\omega$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	40	0
∞	0	-270

Meeting point on imaginary axis, $\angle G(j\omega) = -\tan^{-1} 2\omega - \tan^{-1} 3\omega - \tan^{-1} 4\omega$

$\omega = 0.1961$ rad/sec;

$$|G(j\omega)| = \frac{40}{\sqrt{1+4\omega^2} \times \sqrt{1+9\omega^2} \times \sqrt{1+16\omega^2}} = \frac{40}{\sqrt{1+4 \times 0.1961^2} \times \sqrt{1+9 \times 0.1961^2} \times \sqrt{1+16 \times 0.1961^2}}$$

$$= 25.254$$

$$G(j\omega) = \frac{1}{(j\omega)^2(1+j\omega T)}$$

$$|G(j\omega)| = \frac{1}{\omega^2 \sqrt{1+\omega^2 T^2}}$$

$$\angle G(j\omega) = -180 - \tan^{-1} \omega T$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	∞	-180°
∞	0	-270°

$\omega=0$



Note: $G(j\omega) = \frac{1}{-\omega^2(1+j\omega T)} = \frac{-(1-j\omega T)}{\omega^2(1+j\omega T) \times (1-j\omega T)} = \frac{-1+j\omega T}{\omega^2(1+\omega^2 T^2)}$

$$= \frac{-1}{\omega^2(1+\omega^2 T^2)} + j \frac{T}{\omega(1+\omega^2 T^2)}$$

When $\omega = 0$, both real part and imaginary part are ∞ .

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Polar plot of $G(s) = \frac{1}{s(1+sT_1)(1+sT_2)}$

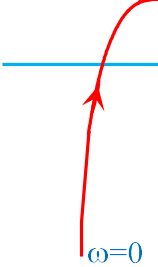
$$G(j\omega) = \frac{1}{(j\omega)(1+j\omega T_1)(1+j\omega T_2)}$$

$$|G(j\omega)| = \frac{1}{\omega \sqrt{1+\omega^2 T_1^2} \times \sqrt{1+\omega^2 T_2^2}}$$

$$\angle G(j\omega) = -90 - \tan^{-1} \omega T_1 - \tan^{-1} \omega T_2$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	∞	-90°
∞	0	-270°

$\omega=0$



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Polar plot of $G(s) = \frac{1}{s^2(1+sT_1)(1+sT_2)}$

$$G(j\omega) = \frac{1}{(j\omega)^2(1+j\omega T_1)(1+j\omega T_2)}$$

$$G(j\omega) = \frac{10}{j\omega(1+j\omega)(1+j0.5\omega)}$$

$$|G(j\omega)| = \frac{10}{\omega\sqrt{1+\omega^2} \times \sqrt{1+(0.5\omega)^2}}$$

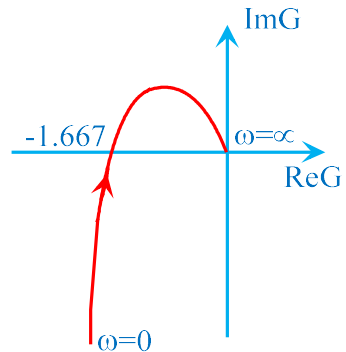
$$\angle G(j\omega) = -90 - \tan^{-1} \omega - \tan^{-1} 0.5\omega$$

ω	$ G(j\omega) $	$\angle G(j\omega)$
0	∞	-90
∞	0	-270

Meeting point on negative real axis, $\angle G(j\omega) = -90 - \tan^{-1} \omega - \tan^{-1} 0.5\omega = -180$

$$\omega = \sqrt{2} ;$$

$$|G(j\omega)| = \frac{5}{\omega\sqrt{1+\omega^2} \times \sqrt{1+(0.5\omega)^2}} = \frac{5}{\sqrt{2} \times \sqrt{1+2} \times \sqrt{1+0.5^2 \times 2}} = 1.667$$

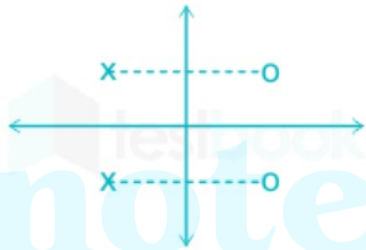


MINIMUM PHASE TRANSFER FUNCTION

A transfer function $G(s)$ is minimum phase if both $G(s)$ and $1/G(s)$ are causal and stable. A minimum phase transfer function has all its poles and zeros in the left-half of the s -plane. A minimum phase system does not have zeros or poles on the right-half s -plane or on the $j\omega$ axis, Moreover, it does not have delay.

$$|G(j\omega)| = \left| \frac{j\omega}{(j\omega + a)} \times \frac{j\omega}{(j\omega + b)} \right| = 1$$

$$\angle G(j\omega) = -2 \tan^{-1} \frac{\omega}{a} - 2 \tan^{-1} \frac{\omega}{b}$$



TRANSPORTATION LAG

The transportation lag is the delay between the time an input signal is applied to the system reacts to that input signal. Transportation lags are common in industrial processes and are often called “dead time”.

Time delay systems are of non-minimum phase behaviour.



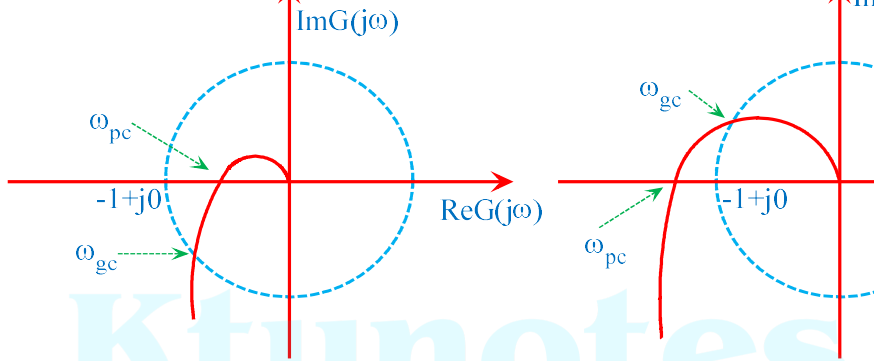
The transfer function of the transportation lag is given by $G(s) = e^{-sT}$, where

$$G(j\omega) = e^{-j\omega T}$$

$$|G(j\omega)| = |e^{-j\omega T}| = 1$$

Introduction of transportation lag has no effect on magnitude plot.

$$\angle G(j\omega) = -\omega T \text{ radians} = -57.3\omega T^\circ$$

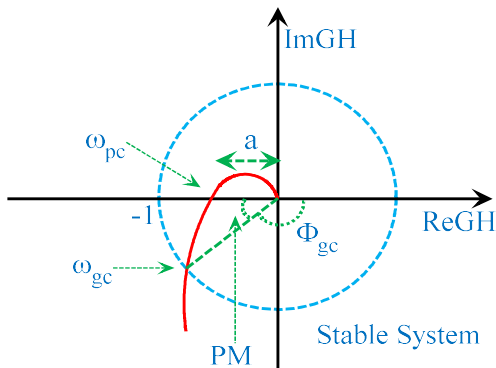


At the point of intersection of polar plot with negative real axis gives the point 180° and the corresponding frequency is ω_{pc} .

The point $-1+j0$ on the negative real axis is called critical point.

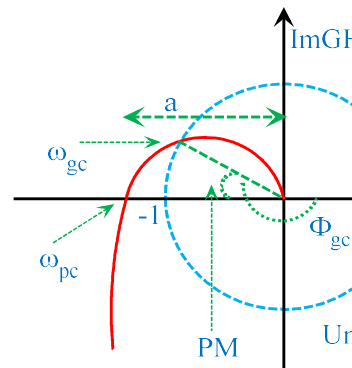
GAIN MARGIN & PHASE MARGIN

As the polar plot gets closer to $-1+j0$ point, the relative stability reduces and the system approaches instability. In frequency domain, relative stability is characterised by gain margin



$$GM = -20\log_{10}a; a < 1; GM \text{ positive}$$

$$PM = 180 + \Phi_{gc}; PM \text{ is positive}$$



$$GM = -20\log_{10}a; a > 1; GM \text{ negative}$$

$$PM = 180 + \Phi_{gc}; PM \text{ is negative}$$

cross-over frequency ω_{gc} to drive the system to the verge of instability.

The value of phase margin can be obtained from the following relation,

$$PM = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc}) = 180^\circ + \phi_{gc}$$

For a stable system, the phase margin is positive.

The critical point $-1+j0$ is outside the polar plot for stable system whereas it is inside for unstable system. If the critical point is enclosed by the polar plot, the system is unstable. If the critical point is not enclosed by the polar plot, the system is stable. (Note: A point will be enclosed by the polar plot if it is on the right-hand side during travelling along the polar plot from the point corresponding to $\omega = 0$ to $\omega = \infty$)

If $\omega_{gc} < \omega_{pc}$, system is stable.

If $\omega_{gc} = \omega_{pc}$, system is marginally stable.

If $\omega_{gc} > \omega_{pc}$, system is unstable.

Note: For non-minimum phase systems, a system may be stable even when the critical point $-1+j0$ is to the left of $-1+j0$ point and thus a negative gain margin may still correspond.

Sketch the polar plot for the open loop transfer function $G(s) = \frac{75}{(s+3)^3(s+1)}$ and find the gain margin and phase margin.

$$G(s) = \frac{75}{(s+3)^3(s+1)} = \frac{75}{27(1+0.333s)^3(1+s)} = \frac{2.778}{(1+0.333s)^3(1+s)}$$

$$G(j\omega) = \frac{2.778}{(1+j0.333\omega)^3(1+j\omega)}$$

$$|G(j\omega)| = \frac{2.778}{(1+(0.333\omega)^2)^{3/2} \times \sqrt{1+\omega^2}}$$

$$\angle G(j\omega) = -3 \times \tan^{-1} 0.333\omega - \tan^{-1} \omega$$

Magnitude at $\omega = 0.876$ rad/sec;

$$|G(j\omega)| = \frac{2.778}{\left(1 + (0.333 \times 0.876)^2\right)^{3/2} \times \sqrt{1 + 0.876^2}} = 1.85$$

Meeting point on positive imaginary axis, $\angle G(j\omega) = -3 \times \tan^{-1} 0.333\omega - \tan^{-1} \omega = 5.941$ rad/sec;

Magnitude at $\omega = 5.941$ rad/sec;

$$|G(j\omega)| = \frac{2.778}{\left(1 + (0.333 \times 5.941)^2\right)^{3/2} \times \sqrt{1 + 5.941^2}} = 0.06$$

Meeting point of polar plot on unit circle,

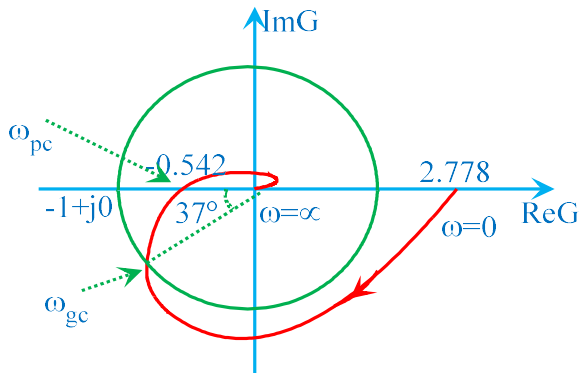
$$|G(j\omega)| = \frac{2.778}{\left(1 + (0.333\omega)^2\right)^{3/2} \times \sqrt{1 + \omega^2}} = 1$$

$\omega_{gc} = 1.614$ rad/sec;

$$\angle G(j\omega_{gc}) = -3 \times \tan^{-1} 0.333\omega_{gc} - \tan^{-1} \omega_{gc} = -3 \times \tan^{-1} 0.333 \times 1.614 - \tan^{-1} 1.614 = -143^\circ$$

Phase Margin, PM = $180 + (-143) = 37^\circ$

Both GM and PM are positive. Hence, the system is stable.



Meeting point on negative real axis, $\angle G(j\omega) = -90 - \tan^{-1} \omega - \tan^{-1} 2\omega = -169^\circ$

$\omega_{pc} = 0.707 \text{ rad/sec}$;

$$|G(j\omega_{pc})| = \frac{1}{\omega_{pc} \sqrt{1 + \omega_{pc}^2} \times \sqrt{1 + 4\omega_{pc}^2}} = \frac{1}{0.707 \times \sqrt{1 + 0.707^2} \times \sqrt{1 + 4 \times 0.707^2}}$$

Gain Margin, $GM = -20 \log_{10} |G(j\omega_{pc})| = 3.517 \text{ dB}$

Meeting point of polar plot on unit circle,

$$|G(j\omega)| = \frac{1}{\omega \times \sqrt{1 + \omega^2} \times \sqrt{1 + 4\omega^2}} = 1$$

$\omega_{gc} = 0.5716 \text{ rad/sec}$;

$$\angle G(j\omega_{gc}) = -90 - \tan^{-1} \omega_{gc} - \tan^{-1} 2\omega_{gc} = -169^\circ$$

Phase Margin, $PM = 180 + (-169) = 11^\circ$

Both GM and PM are positive. Hence, the system is stable.

